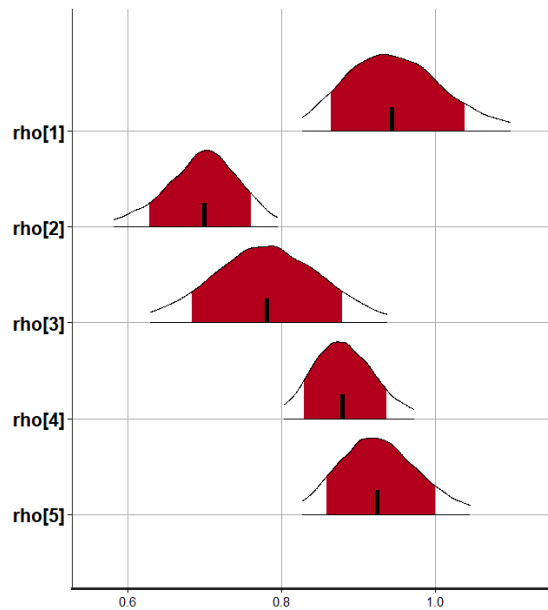


Question #1.

(1)



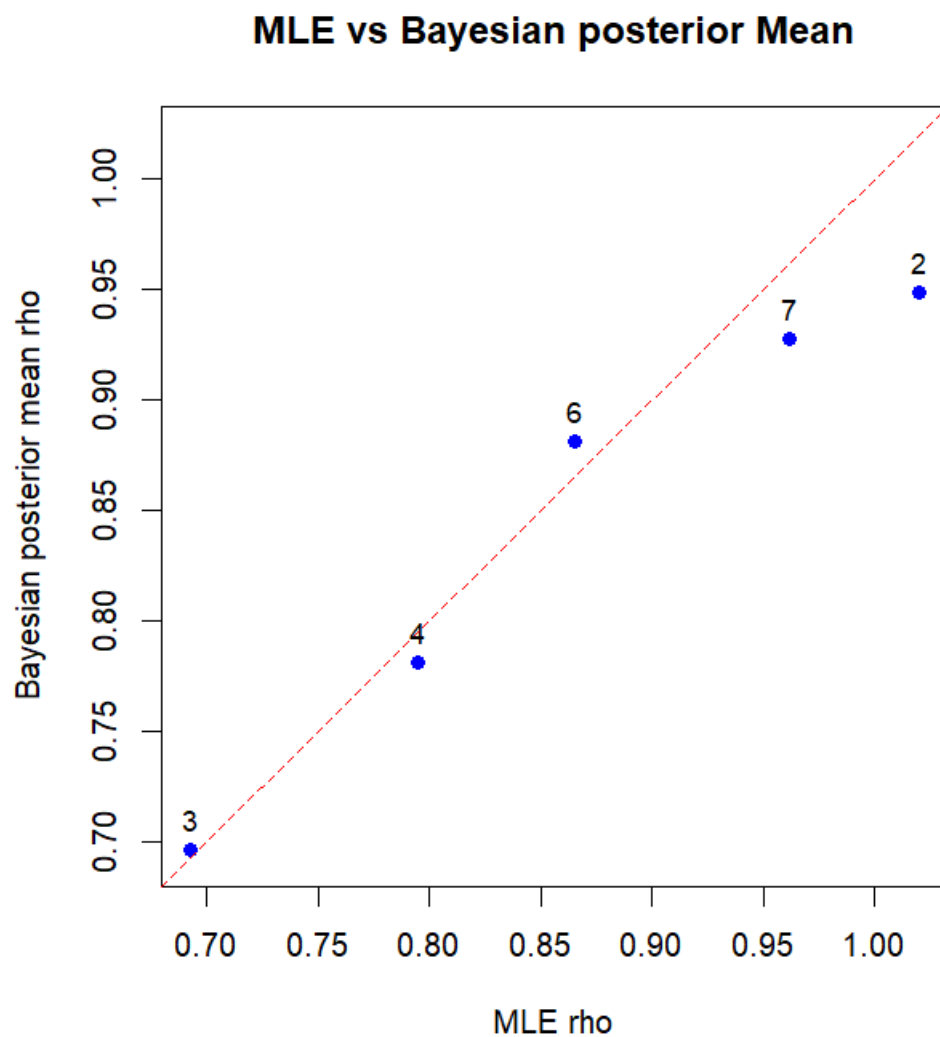
ci_level: 0.8 (80% intervals) outer_level: 0.95 (95% intervals)

Bayesian	subjID	rho_postmean
1	2	0.9488431
2	3	0.6962745
3	4	0.7812755
4	6	0.8816324
5	7	0.9279043

MLE(from HW 2)	Subject	rho
1	2	1.019878
2	3	0.692663
3	4	0.794402
4	6	0.865562
5	7	0.961615

I selected rho because it directly shows the degree of risk aversion, which makes easier explanation for difference between the subjects.

(2)



(3) MLE and hierarchical Bayesian estimation produced very similar parameter estimates overall, but hierarchical Bayesian estimation combines individual and group level information, which makes shrinkage. Therefore the result of hierarchical Bayesian estimation seems to reduce extreme estimates such as Sample 2(1.019878 -> 0.9488431) to the posterior means of the group.

Question #2.

a) For calculating Bayes factor, we should know marginal likelihood. But calculating marginal likelihood is difficult, so we can use BIC as an alternative option. Marginal likelihood and BIC has this relationship :

$$\log P(y|M) \approx -0.5 \times BIC$$

Because Bayes factor is $P(y|M1) / P(y|M2)$, it can be calculated by getting log :

$$\text{Log}(\text{BF}_{12}) = \text{Log}(P(y|M1)) - \text{Log}(P(y|M2)) = -0.5 \times (\text{BIC1} - \text{BIC2})$$

Also, BIC can be calculated as this :

$\text{BIC} = -2\log L + k\log(n)$, and k = parameter numbers = 3(ra_prospect) or 2(ra_noLA), n = trial number = 140

Therefore each BIC value and Bayes factor is calculated like this :

1) ra_prospect

Subject	minusLL	BIC1
2	68.67925	143.7969
3	18.46703	43.37244
4	81.01863	168.4756
6	43.79465	94.02768
7	41.19514	88.82866

2) ra_noLA

Subject	minusLL	BIC2
2	74.48128	153.254816
3	43.27384	90.8399361
4	81.40108	167.094416
6	43.89129	92.0748361
7	58.1771	120.646456

3) Bayes factor

Subject	BIC1	BIC2	log(Bayes Factor)	Bayes factor
2	143.7968841	153.254816	4.72896598	53575.46908
3	43.37244411	90.8399361	23.733746	5.41684E+23
4	168.4756441	167.094416	-0.690614	0.203885332

6	94.02768411	92.0748361	-0.976424	0.10557862
7	88.82866411	120.646456	15.908896	8.10767E+15

b) According to Jefferey's evidence categories(modified by Lee and Wagenmakers (2014)), Bayes factor means the grades of evidence like this :

$B_{1,2}$	Grades of evidence
1 to 3	Anecdotal
3 to 10	Moderate
10 to 30	Strong
30 to 100	Very strong
> 100	Extreme

Therefore, for subject 4 and 6, the model 1(ra_prospect) showed anecdotal(week) evidences compared to model 2(ra_noLA). However, for the other subjects, the model 1 showed extreme evidences.

c) posterior odds = prior odds * Bayes factor, so it is $(0.6/0.4) \times \text{Bayes factor}$.

Therefore, posterior odds for each subjects are like this :

Subject	Bayes factor	posterior odds
2	53575.4691	80363.20361
3	5.4168E+23	8.12526E+23
4	0.20388533	0.305827998
6	0.10557862	0.15836793
7	8.1077E+15	1.21615E+16

d) Now we are calculating the posterior model probability of model 1, which means posterior probability of M1 is like this : $P(M1|y) = PO12 / (1+PO12)$

Therefore, the answer is like this :

Subject	Posterior odds	Posterior probability of model 1
2	80363.20361	0.99998755
3	8.13E+23	1
4	0.305827998	0.2339649

6	0.15836793	0.13668418
7	1.22E+16	1

e) In HW2, I got a BIC value for each model like this :

	Models	AIC	BIC
1	POW1	475.213	475.292
2	POW2	477.213	477.371
3	EXP1	524.452	524.532
4	EXP2	489.176	489.335
5	EXPOW	479.213	479.451
6	HYP1	482.875	482.954
7	HYP2	481.475	481.634

As question 2a, I approximated the Bayes Factor(Each model is model1, and EXP1 is set to model 2 due to the highest BIC)

	Models	AIC	BIC	BIC2	log(Bayes Factor)	Bayes factor
1	POW1	475.213	475.29	475.292	0	1
2	POW2	477.213	477.37	475.292	-1.0395	0.09130614
3	EXP1	524.452	524.53	475.292	-24.62	2.3988E-25
4	EXP2	489.176	489.34	475.292	-7.0215	9.517E-08
5	EXPOW	479.213	479.45	475.292	-2.0795	0.00832722
6	HYP1	482.875	482.95	475.292	-3.831	0.00014757
7	HYP2	481.475	481.63	475.292	-3.171	0.00067453

Therefore, POW1 shows the highest Bayes factor – which means that it is the most powerful model among 7 models, but POW2 can also be a good alternative because it has second highest Bayes factor.

Question #3.

a) Yes, I put for loop within Stan and changed the data type from scalar to vector.

b)

Posterior means of rho by subject	subjID	rho_mean
1	2	1.030704
2	3	0.680756
3	4	0.791087
4	6	0.868629
5	7	0.958725